B.TECH. DEGREE V SEMESTER EXAMINATION, Nov 2024
19-205-0503 Machining Science and Machine Tools

Time: 3 Hours
Marks: 60

CO1	To study the geometry of single point and multi point cutting tools and mechanics of chip formation
CO2	To get the concept of mechanics of metal cutting
CO3	To understand the principles of working, specifications, different types and operations performed in lathe, drilling and shaping machines
CO4	To learn the principles of working, specifications, different types and operations performed in grinding and milling machines

Bloom's taxonomy levels (L1-Remember, L2-Understand L3-Apply, L4-Analyze L5-Evaluate L6-Create)
PO-Program outcome

PART A
Answer all questions
(8x3=24)

I		Marks	BL	CO	PO
a	Explain the formation of a built-up-edge (BUE). Explain the conditions which promote the growth of BUE.	3	L2	1	1
b	How does the cutting process parameters affect the cutting tool wear in single point tools?	3	L2	2	1
c	Compare 'generating' and 'forming' with reference to metal cutting machine tools?	3	L3	3	1
d	What is the function served by a lead screw in a machine tool?	3	L1	3	1
e	How a lathe is specified?	3	L1	3	1
f	Describe Indian Standard wheel marking system for grinding wheels	3	L1	4	1
g	Differentiate between up milling and down milling	3	L2	4	1
h	What are the essential requirements of clamps in fixture design?	3	L2	4	1

PART B
(12*4=48)

II		Marks	BL	CO	PO
	(a) Explain the flank wear of a cutting tool with the help of suitable sketches. Also explain its importance from the tool		L2	2	1

	life point of view. (b) Find the power consumption in terms of shear and friction angles, during machining process. Use Merchant's circle diagram for derivation.	5 7	L4	2	2
	OR				
III	(a) During an orthogonal machining operation on mild steel, the results obtained are: t_1 = 0.25mm, t_2 = 0.75mm, w = 2.5mm, α= 0°, Fc = 950N and Ft= 475N. (i) Determine the coefficient of friction between the tool and the chip and (ii) the ultimate shear stress of the work material.	6	L3	2	2
	(b) Explain the major criteria for judging machinability.	3	L1	2	1
	(c) Explain the relationship of tool temperature and tool life.	3	L1	2	1
IV	(a) What are auxiliary motions in a machine tool? Give examples.	5	L2	3	1
	(b) Briefly explain the ratchet-gear mechanism used for intermittent motions in machine tools with the help of sketches.	7	L1	3	1
	OR				
V	(a) Explain the application of differential mechanism in machine tools.	5	L2	3	1
	(b) Explain the function of a translatory hydraulic drive in machine tools.	7	L2	3	1
VI	(a) What are the different methods used for taper turning? Explain any two.	7	L2	3	1
	(b) A cast iron workpiece of 50mm diameter and 200 mm length is rough turned on lathe with a tool of lead angle 20 and cutting speed of 70 m/min. Feed rate is 0.4mm/rev. and depth of cut is 4mm. Determine the material removal rate and machining time.	5	L3	3	2
	OR				
VII	(a) With a neat sketch explain the crank and slotted link quick return mechanism of shaper	7	L1	3	1
	(b) A plate 600 X 600 mm is to be machined on a shaper. Find the time required for a complete cut, if the cutting speed is 9 m/min. The return time to cutting time ratio is 1: 4	5	L3	3	2

	and the feed is 2 mm. The clearance at each end is 75 mm.				
VIII	With a neat sketch explain the geometry of milling cutter	6	L1	4	1
	(b) Determine the time required to mill a slot in a work piece of 490 mm length with a side milling cutter of 100 mm diameter, 25 mm wide and 18 teeth. The depth of cut is 5mm and feed per tooth is 0.1 mm. Cutting speed used is 30 m/min. Assume a clearance of 2.5 mm.	6	L3	4	2
	OR				
IX	**(a)** What are the factors to be considered for the selection of grinding wheels?	6	L2	4	1
	(b) Explain the features of an indexing jig. Give sketches.	6	L2	4	1

L1=29.17% L2=44.16% L3=20.84% L4=5.83%